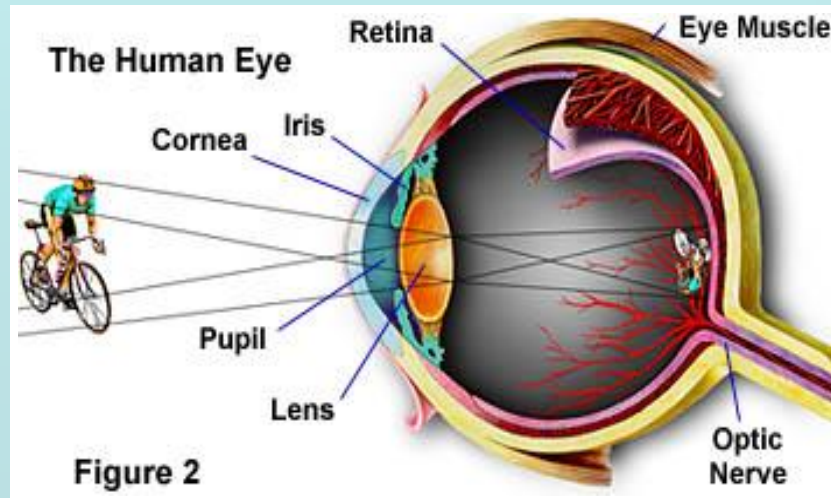
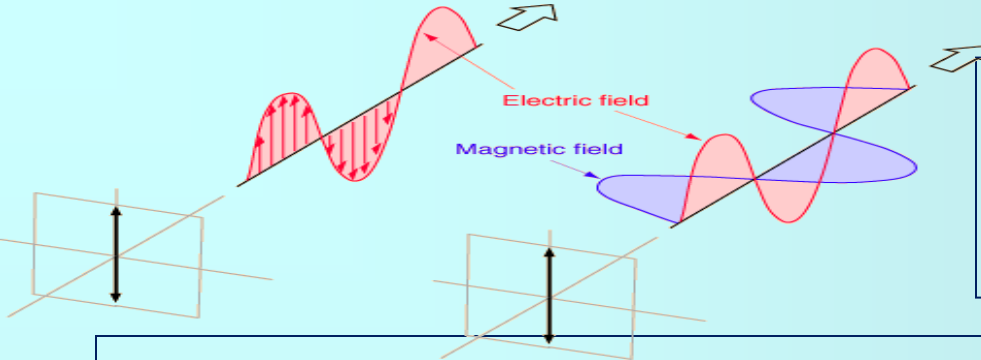


LIGHT

LIGHT

- Light is an invisible energy which causes in us the sensation of sight (vision)
- Light is an agent which stimulates our sense of sight
- The light waves enter the retina of the eye, and stimulate parts of it that cause a electrical impulse to be sent to the brain which creates this visual image.
- Light is propagated in the form of electromagnetic waves like radio waves and microwaves travelling at speed of $3 \times 10^8 \text{ m/s}$





Rectilinear propagation of light

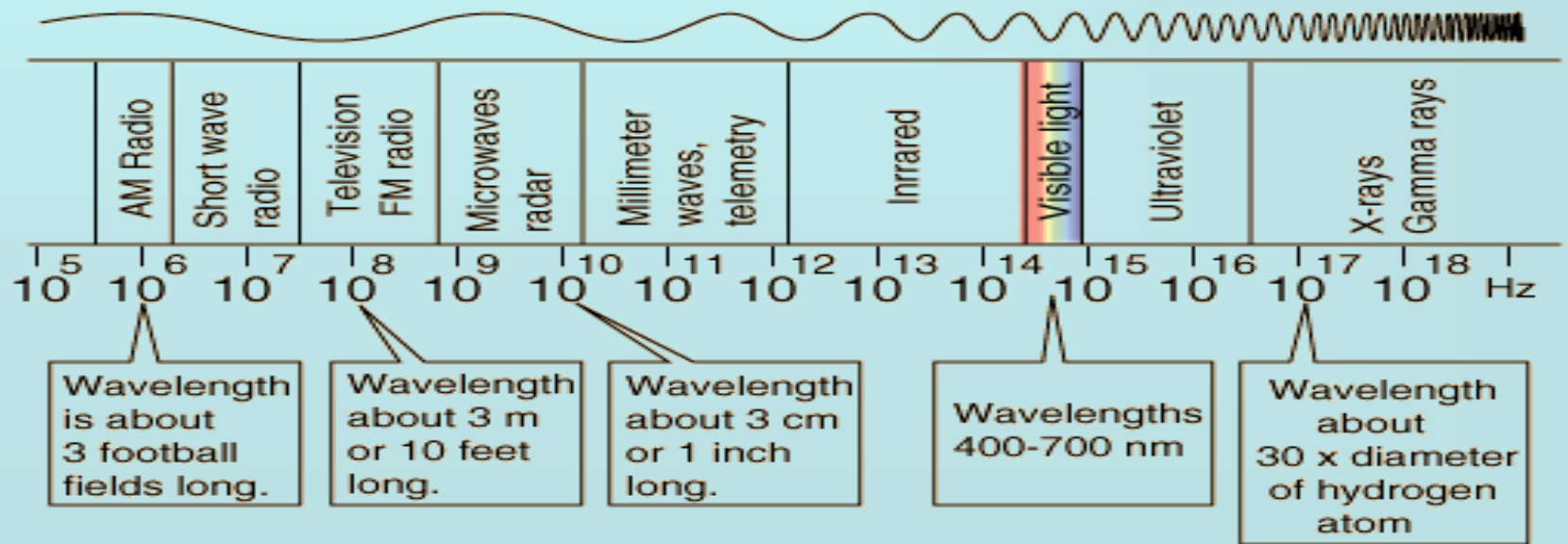
- Light energy consists of tiny packets of energy called photons
- Each photon has an electric field and a magnetic field around it
- These fields mutually at right angles to each other
- Due to the interaction of above fields, the photon moves mutually at right angles to both the fields i.e, in a straight line
- This kind of propagation of light photons is called rectilinear propagation of light (light energy travels in straight lines)

Electromagnetic Waves

- Electrons in materials are vibrated and emit energy in the form of photons, which propagate across the universe
- Photons have no mass, but are pure energy
- Electromagnetic Waves are waves that are made up of these “photons”
- When these photons come in contact with boundaries, E-M waves interact like other waves would
- Electromagnetic waves are everywhere, Light is only a small part of them

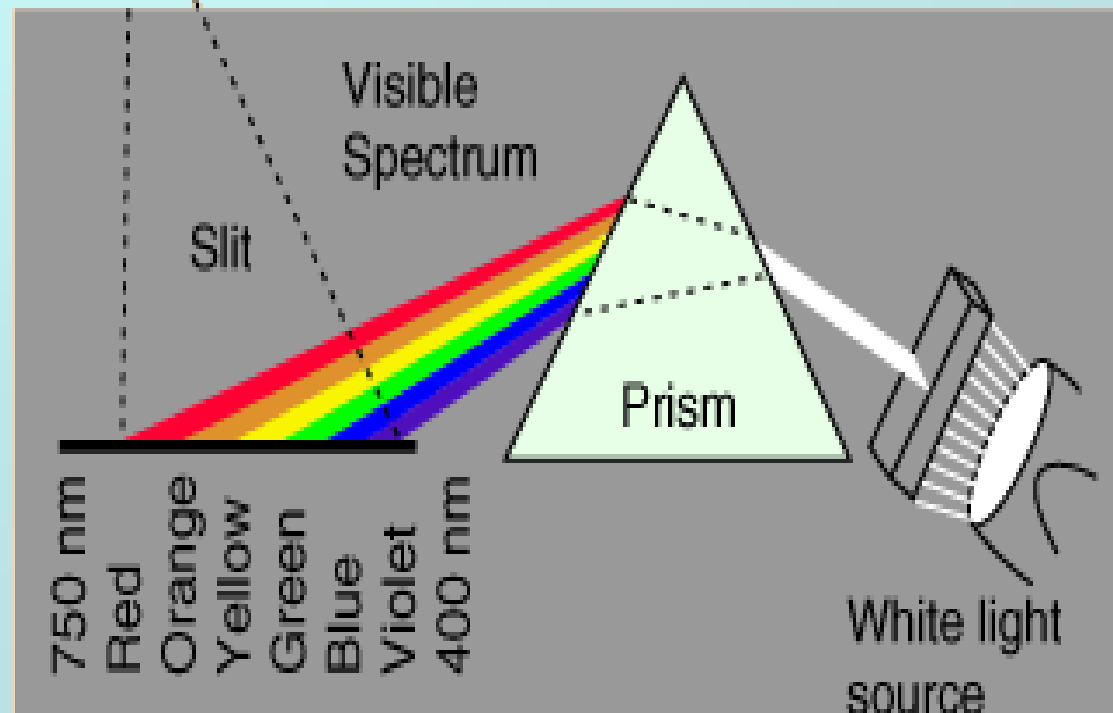
Electromagnetic spectrum

- Only those waves within a certain band of frequencies which can enter the eye and produces the sensation of light, are called light rays
- The other electromagnetic waves are invisible.



Visible spectrum

Radio	Far IR, Micro-wave	IR	UV	x-ray γ -ray
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Radiation	Wavelength	Photon energy
Radio, TV, Radar	$3 \times 10^4 \text{ m} - 100 \text{ }\mu\text{m}$	$4.1 \times 10^{-11} - 1.2 \times 10^{-2} \text{ eV}$
Infra red	$100 \text{ }\mu\text{m} - 700 \text{ nm}$	$1.2 \times 10^{-2} - 1.8 \text{ eV}$
Visible light	$700 - 400 \text{ nm}$	$1.8 - 3.1 \text{ eV}$
Ultra violet	$400 - 10 \text{ nm}$	$3.1 - 124 \text{ eV}$
X- rays & gamma	$10 - 10^{-4} \text{ nm}$	$124 - 124 \times 10^6 \text{ eV}$
Cosmic rays	$10^{-4} - 10^{-7} \text{ nm}$	$10^7 - 10^{11} \text{ eV}$

General definitions pertaining to light

- **Luminous bodies** : luminous bodies are those which emit light energy by themselves. e.g sun, stars, burning candle, glowing electric bulb, glow worm, certain kinds of fish
- **Non-luminous bodies**: The bodies which do not give light energy of their own, but reflect light energy falling on them. e.g wood, house, furniture, book etc

Transmission of light

Based on transmission of light, the substances can be divided in to three categories namely:

1. Transparent
2. Translucent
- 3.opaque

Definition and example

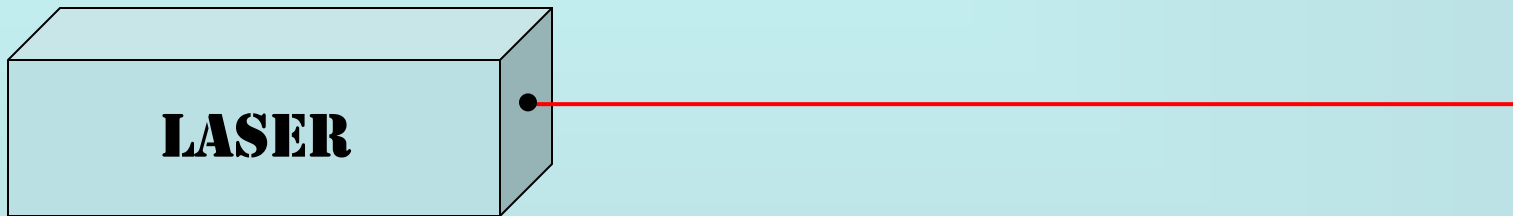
- **Transparent** : A medium which allows most of the light energy to pass through it, is called transparent.e.g vacuum, clear air, thin layers of water or alcohol, glass, certain kinds of plastics and diamond
- **Translucent**: A medium which partially allows the light energy to pass through it, is called translucent medium. e.g butter paper, oiled paper, tissue paper, ground glass, mist
- **Opaque** : opaque substances are those which do not allow light to pass through them.e.g bricks, wood, stones, metal

Properties of Light

- 1) Light travels in straight lines
- 2) Light travels much faster than sound
- 3) We see things because they reflect light into our eyes
- 4) Shadows are formed when light is blocked by an object

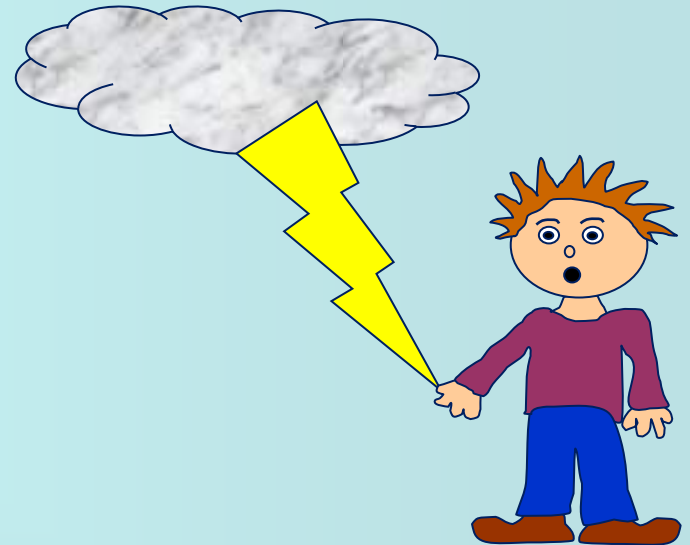
Properties of Light

- Light travels in straight lines

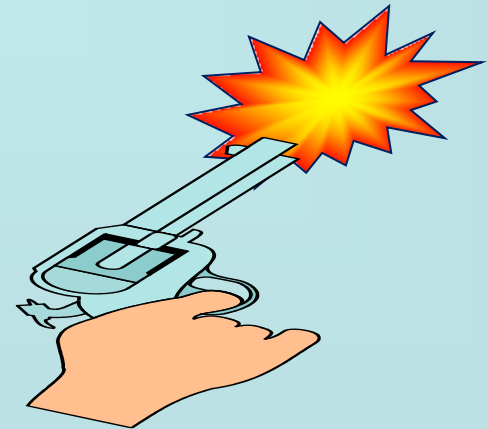


Light travels much faster than sound. For example:

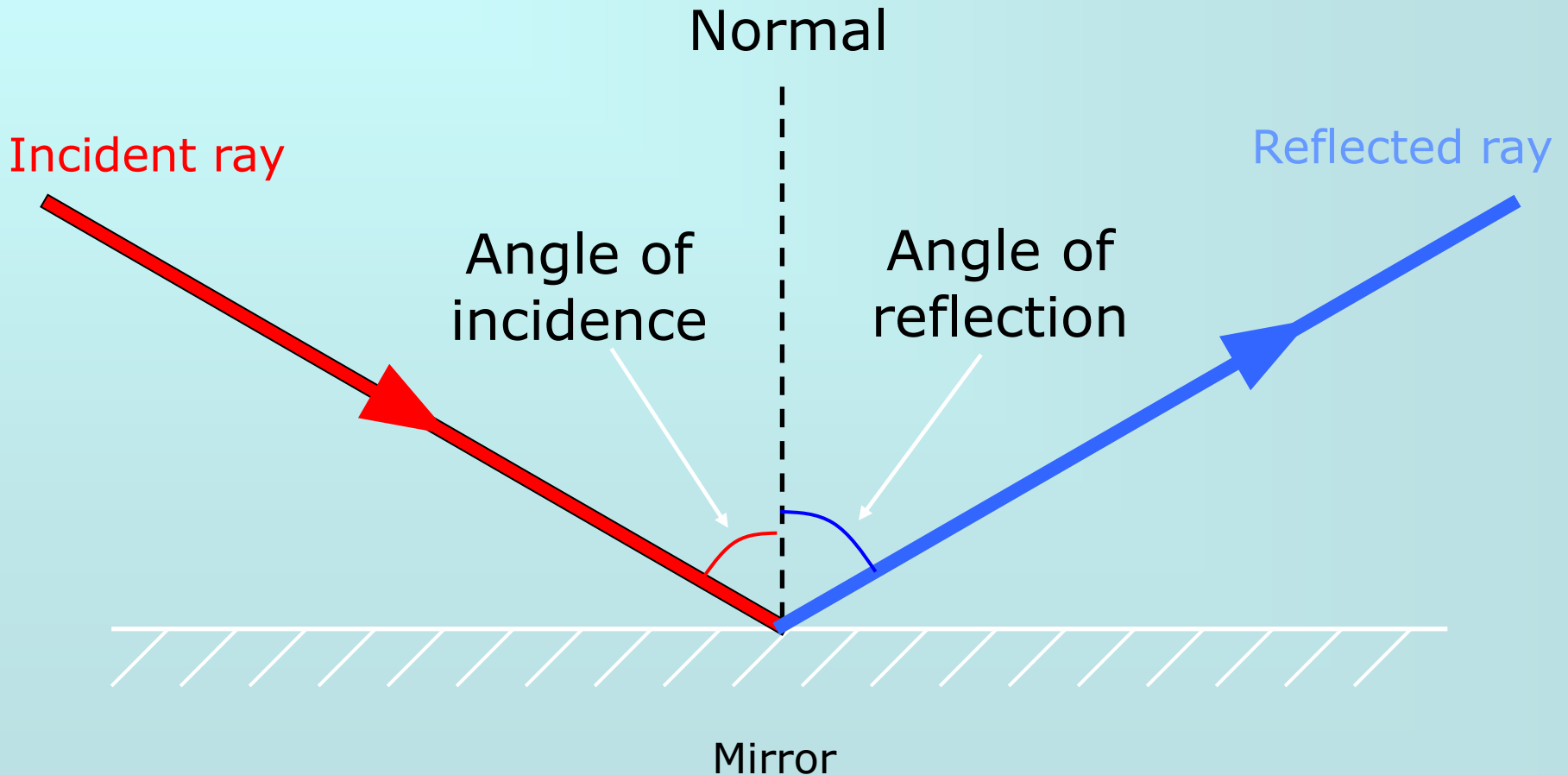
1) Thunder and lightning start at the same time, but we will see the lightning first.



2) When a starting pistol is fired we see the smoke first and then hear the bang.

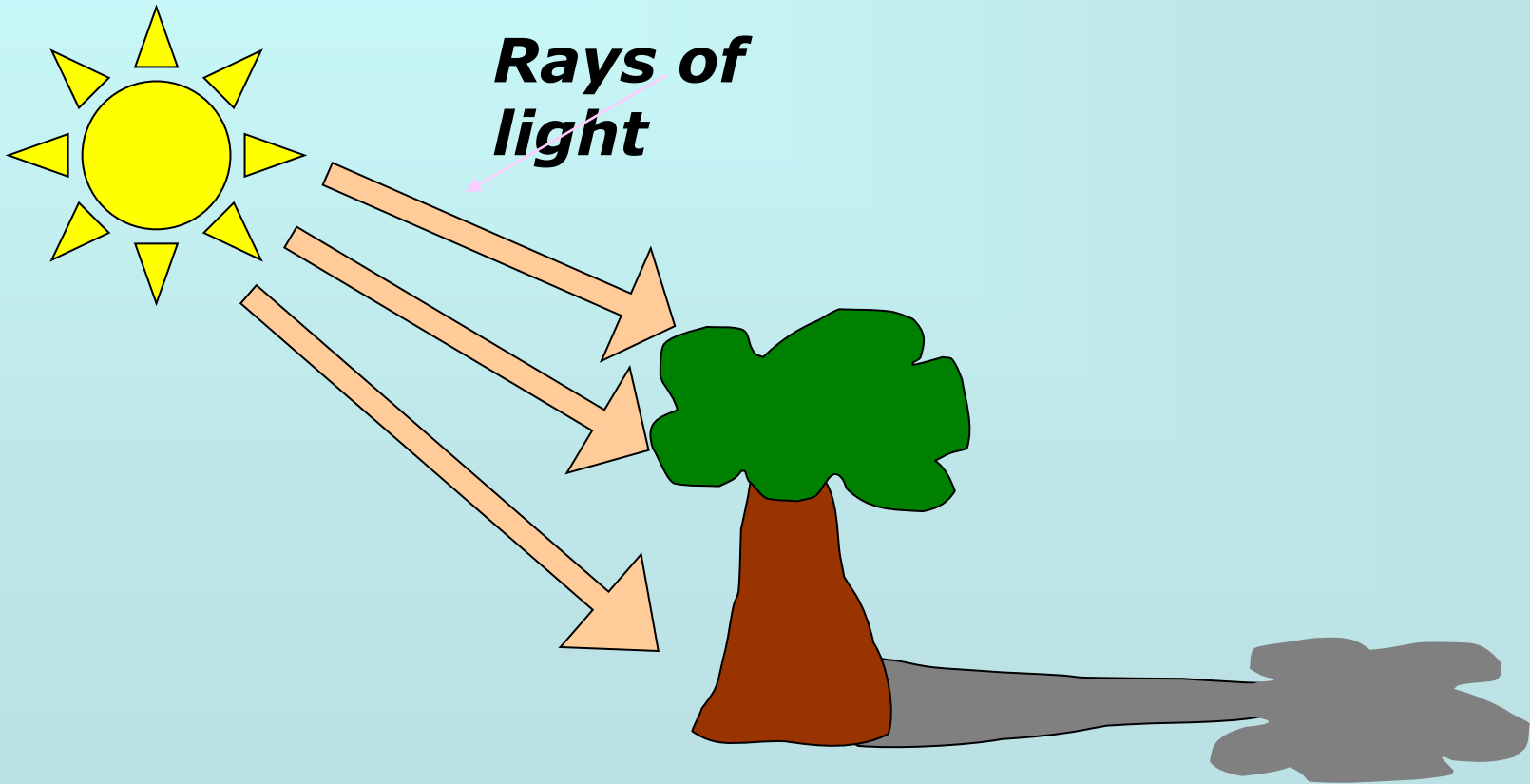


We see things because they reflect light into our eyes - Reflection from a mirror



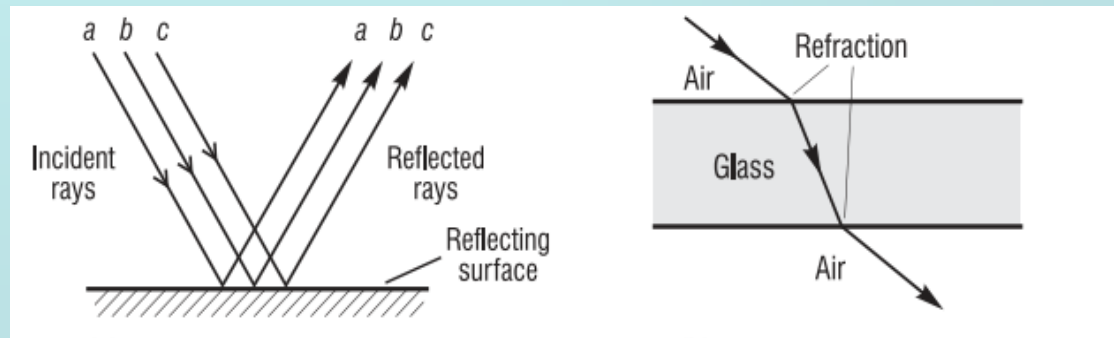
Shadows

Shadows are places where light is “blocked”:



Reflection and refraction

1. A part of light, on striking the surface, is sent back to the original medium. It is called reflection of light
2. A part of light, on striking the surface gets transmitted through it. It is called refraction of light.



Reflection

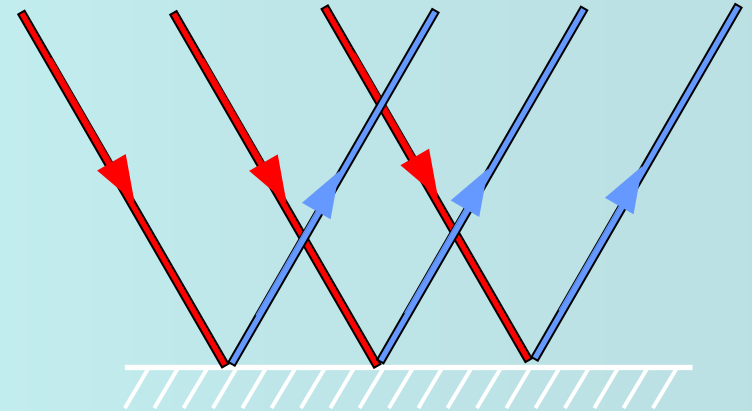
- The phenomenon due to which a beam of light, travelling through certain medium, on striking some other surface, bounces off from it in some other direction is called reflection of light.

Reflection-types

- **Regular reflection** : It occurs when a parallel beam of light falls on a smooth and polished surface like a plane mirror.
- **Irregular reflection** : It occurs when a parallel beam of light strikes a rough and unpolished surface. Each ray of the beam is reflected in a different direction.

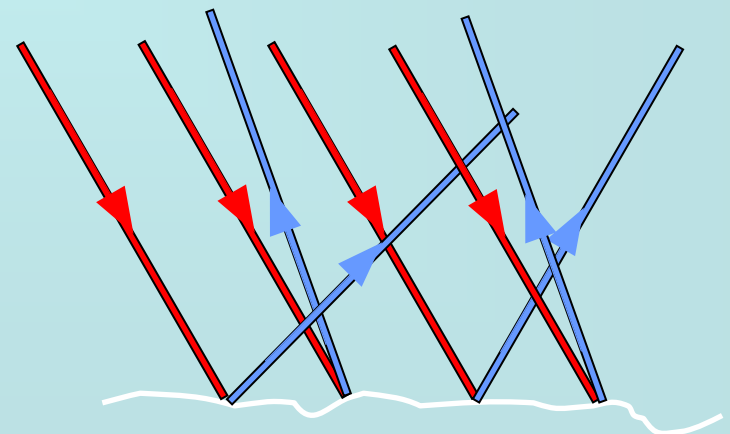
Clear vs. Diffuse Reflection

Smooth, shiny surfaces have a clear reflection



Rough, dull surfaces have a *diffuse* reflection.

Diffuse reflection is when light is scattered in different directions



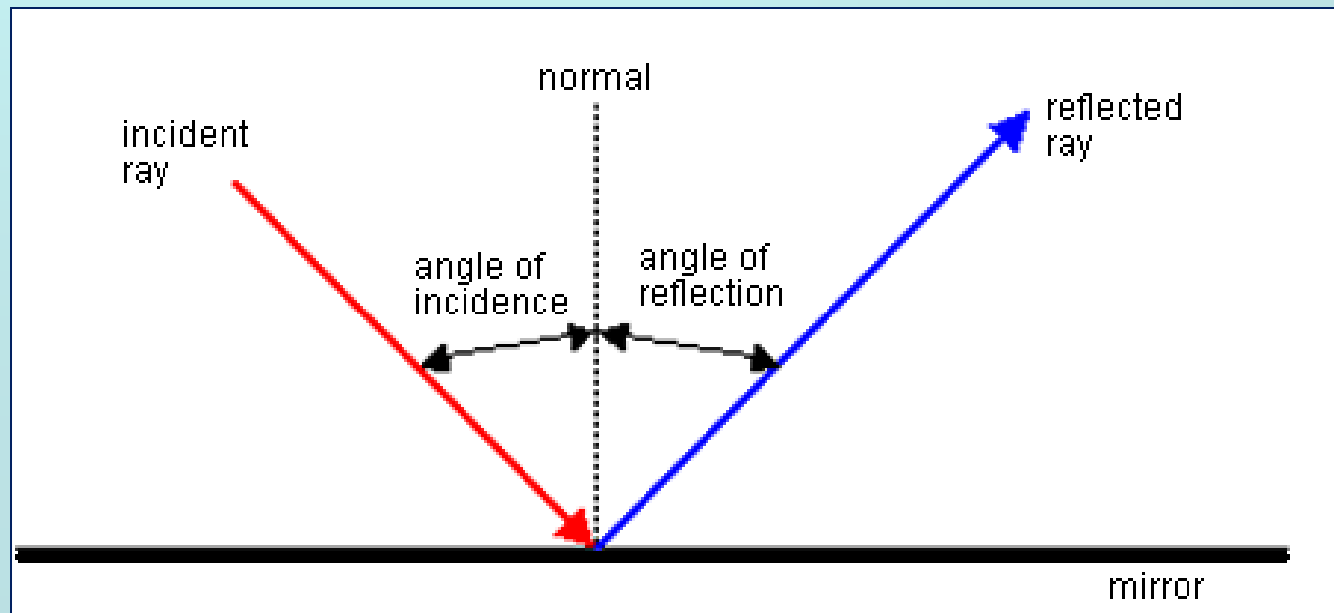
Reflection of light from optical surfaces

When light is incident on an *interface between two transparent optical media*—such as between air and glass or between water and glass—four things can happen to the incident light

- It can be partly or totally reflected at the interface.
- It can be scattered in random directions at the interface.
- It can be partly transmitted via refraction at the interface and enter the second medium.
- It can be partly absorbed in either medium.

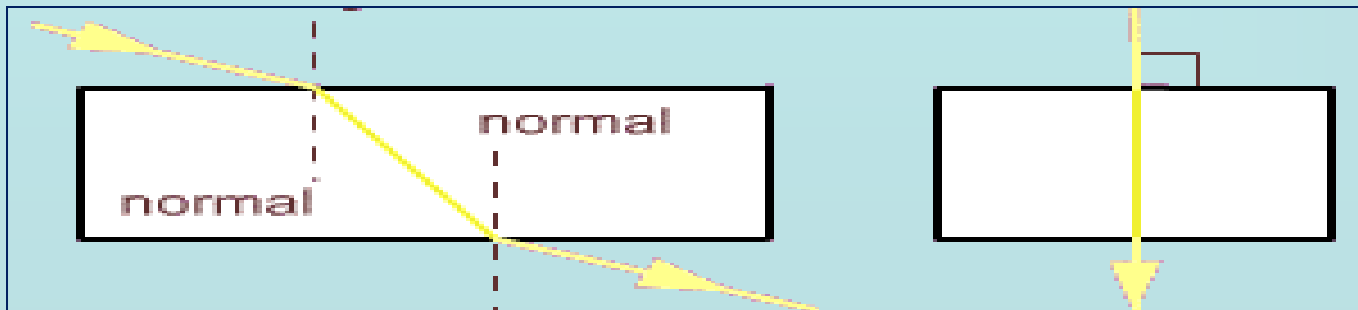
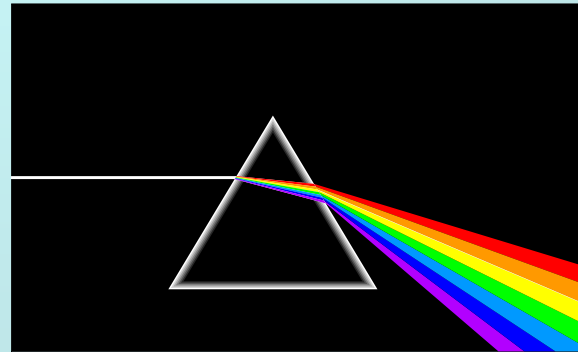
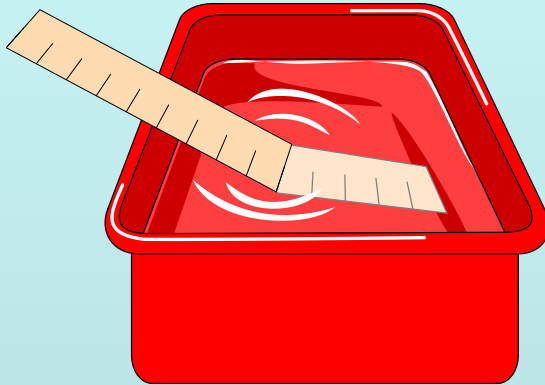
Laws of reflection

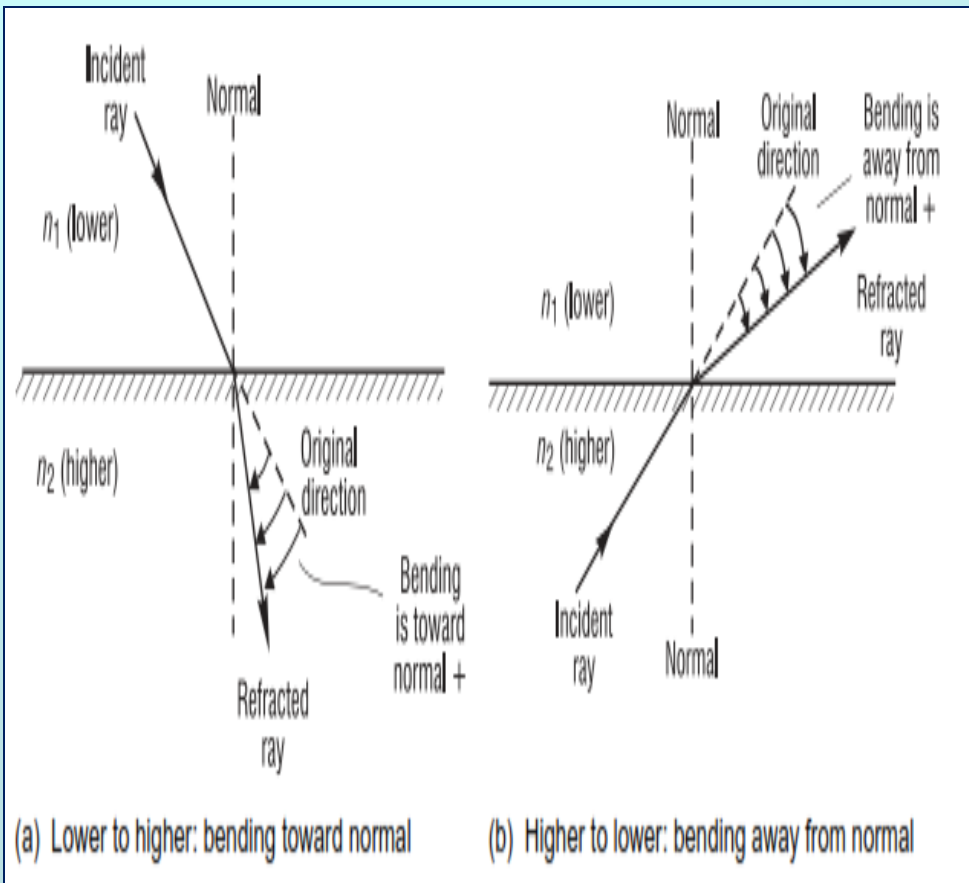
1. The incident ray, the normal and the reflected ray at the point of incidence, all lie in the same plane.
2. The angle of incidence is equal to the angle of reflection.



Refraction

- Bending of light, when it travels from one medium to other is called refraction of light.





- When a ray of light proceeds from a rarer to a denser medium, it bends towards the normal, as its velocity decreases.
- When a ray of light proceeds from a denser to a rarer medium, it bends away from the normal, as its velocity increases.

Laws of refraction

1.The incident ray, the normal and the refracted ray drawn at the point of incidence, all lie in the same plane.

2.snell's law : The ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant quantity which is called refractive index of the second medium with respect to the first.

$$\sin i / \sin r = {}_1\mu_2 = {}_1n_2$$

Index of refraction

- The two transparent optical media that form an interface are distinguished from one another by a constant called the *index of refraction*, generally labeled with the symbol n .
- *The index of refraction for any transparent optical medium is defined as the ratio of the speed of light in a vacuum to the speed of light in the medium*

$$n = \frac{c}{v}$$

- where c = *speed of light in free space (vacuum)*
- v = speed of light in the medium
- n = *index of refraction of the medium*
- The index of refraction for free space is exactly *one*. For air and most gases it is very nearly one, so in most calculations it is taken to be 1.0. For other materials it has values greater than one.

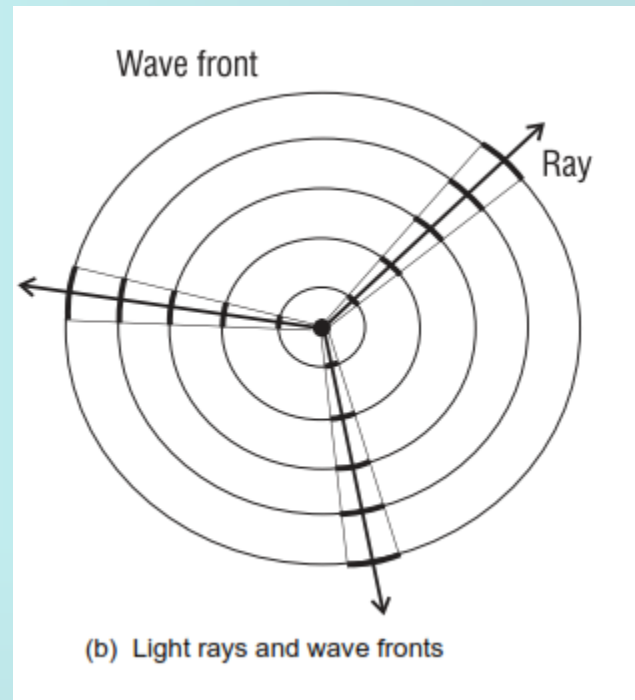
Light rays and light waves

- A light ray is nothing more than an imaginary line directed along the path that the light follows
- It is helpful to think of a light ray as a narrow pencil of light, very much like a narrow, well-defined laser beam
- wave motion is easily visualized in terms of water waves—such as those created on a quiet pond by a bobbing cork.

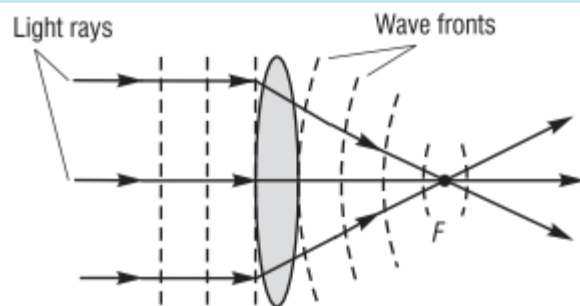
- The successive high points (crests) and low points (troughs) occur as a train of circular waves moving radially outward from the bobbing cork
- Each of the circular waves represents a *wave front*
- *A wave front is defined here as a locus* of points that connect identical wave displacements—that is, identical positions above or below the normal surface of the quiet pond.



(a) Waves from a bobbing cork



(b) Light rays and wave fronts



(c) Changing wave fronts and bending light rays

Wave front

- Consider a point source 'S' of light placed at a point in air or vacuum.
- The point source of light will emit the light waves in all directions with same velocity in the same medium. Let ' v ' be the velocity of light in air medium.
- In time ' t ' all the waves covers a distance equal to ' vt '. In the other words in time ' t ' all the waves will reach on a spherical surface of radius vt and centre S.
- The spherical surface so formed is called wavefront
- It should be noted that all points on the spherical surface should be in the same state of vibration.

Definition of wavefront

“The locus of all the points of the medium to which all the waves will reach simultaneously, so that all the points are in the same state of vibration is called wavefront”

Types of wavefronts

- There are **three** types of wavefronts
 1. Spherical wavefront.
 2. Plane wavefront.
 3. Cylindrical wavefront.

- **Spherical wavefront** : A wavefront originating from the point source of light at finite distance is called spherical wavefront.
- **Plane wavefront** : As time passes the spherical surface of spherical wavefront is so large that a small part of it can be considered as a plane wavefront
 - A wavefront originating from the point source of light at infinite distance i.e. at very large distance is called as plane wavefront.
 - The plane wavefront can be obtained by keeping point source at the focus of the convex lens.
 - The light from the sun reaches the earth in the form of plane wavefronts
- **Cylindrical wavefront** : A wavefront originating from a linear source of light is called cylindrical wavefront.

Wave normal

- A normal or perpendicular drawn to the surface of a wavefront at any point, in the direction of propagation of light, is called a wave normal
- The wave normal gives the direction of propagation of light

THANK YOU